Simulation Analysis

Sophie Litschwartz

2025-05-30

Contents

1 Introduction **1**

2 Research Questions **2**

3 Data Generating Process **2**

4 Are any of the estimators ever substantially biased? When? **6**

5 Are any of the estimators generally more precise than others? When? **7**

6 Which estimators produce the best estimates of the treatment effect standard error? **9**

7 Do any of the estimated standard errors target finite versus super population uncertainty? **15**

8 Are problems caused by low degrees of freedom due to estimate instability or bias? **17**

1 Introduction

Blocked cluster randomized controlled trials (CRTs) are a widely used experimental evaluation design in which entire groups (clusters) such as schools or classrooms are randomly assigned to treatment or control conditions within defined strata or blocks. The effectiveness of the program or treatment is then typically quantified using the average treatment effect (ATE). However, within this relatively straightforward quantity, researchers must make a variety of choices. Researchers need to decide whether to target a person-weighted average or cluster-weighted average; they need to decide whether they are targeting an estimate just for the clusters in the study (finite sample) or an estimate representative of some broader population (super-population); and they need to decide from among different methods or estimators that can be used to estimate the ATE.

This paper contains simulations that complement research examining empirically the implications of different choices researchers make when performing CRTs (CITATION). Simulations, unlike empirical analyses, allow us to compare estimated quantities to their true values, which we cannot observe in empirical contexts. This in turn allows us to evaluate how well different statistical methods perform. Additionally, we can systematically vary data conditions and identify which factors impact method performance.

2 Research Questions

With this in mind, these simulations answer the following research questions:

1. Are any of the estimators ever substantially biased? When?
2. Are any of the estimators generally more precise than others? When?
3. Which estimators produce the best estimates of the treatment effect standard error?
4. Do any of the estimated standard errors target finite versus super-population uncertainty?
5. Are the estimation problems caused by low degrees of freedom due to estimate instability or bias?

3 Data Generating Process

- these are the parameters, but I'm confused on exactly how we're getting the output
- Note I have doubled checked all of these parameters and I'm sure they're correct at this point

The simulations are a fully factored experiment with the following data generating parameters.

Table 1: Simulation Data Generating Parameters

Parameter	**Values**
Block-Level Fixed	
Number of Blocks	5
Average Treatment Effect	.2
Block-Level ICC	.2
Average Number of Observations Per Cluster	20
Number of Block Level Covariates	0
Block-Level Varying	
Block Type	Same, Messy, Reasonable
Average Number of Clusters per Block	5, 10, 20
Proportion of Clusters Treated	.5, .7
Cluster-Level Varying	
Correlation of Impact and Size of Cluster	0, .8
Cluster Level Treatment Effect Standard Deviation	0, .3
Cluster Level ICC	0, .2
How much does the size of clusters vary (coefficient of variation)?	0, .33, .6
Number of Cluster-Level Covariates	0, 1, 5 [R^2 Always .5]

Table 2: Block-Type Parameters

Parameter	**Blocks All The Same**	**Blocks All Messy**	**Blocks All Reasonable**
How much do number clusters per block vary (coefficient of variation)?	0 (none)	.85 (big)	.33 (average)
Block-Level Average Treatment Effect Standard Deviation	0	.6	.2
Correlation Between Treatment Effect and Cluster Size	0	.8	.2
Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters	0	.8	.3
Correlation of Proportion of Clusters Treated per Block and Impact	0	.8	0
Correlation Between Outcome Intercept and Cluster Size	0	.8	.3

We also ran a separate set of simulations to evaluate how standard errors perform for fully interacted models in matched studies. These simulations were also fully factorer and the parameters for those simulations were as follows:

Table 3: Matched Simulation Data Generating Parameters

Parameter	Values
Block-Level Fixed	
Number of Blocks	20
Average Treatment Effect	.2
Block-Level ICC	.2
Average Number of Observations Per Cluster	20
Number of Block Level Covariates	0
Proportion of Clusters Treated	.5
Average Number of Clusters per Block	2
How much do number clusters per block vary (coefficient of variation)?	0
Correlation Between Treatment Effect and Cluster Size	0
Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters	0
Correlation of Proportion of Clusters Treated per Block and Impact	0
Correlation Between Outcome Intercept and Cluster Size	0
Block-Level Varying	
Block-Level Average Treatment Effect Standard Deviation	0, .6, .2
Cluster-Level Fixed	
Number of Cluster-Level Covariates	0
Cluster-Level Varying	
Correlation of Impact and Size of Cluster	0, .8
Cluster Level Treatment Effect Standard Deviation	0, .3
Cluster Level ICC	0, .2
How much does the size of clusters vary (coefficient of variation)?	0, .33, .6

4 Are any of the estimators ever substantially biased? When?

In 90% of the scenarios we tested, the magnitude of bias was less than 0.01 for all estimators (Figure 1), and the scenarios with bias require the confluence of multiple bias-causing factors (Figure 2). The estimators that theory predicts are unbiased do indeed have nearly no bias (or very low bias) across the full range of simulations. The person-weighted fully interacted estimator had bias as severe as -0.033 in the worst 5% of scenarios tested, which the regression tree shows requires a correlation between the treatment effect and cluster size combined with a small number of clusters per block (Figure 2). For the remaining estimators, except the multilevel fully interacted model, we observe bias as severe as -0.2 in some simulations, primarily driven by the combination of dissimilar blocks and unbalanced treatment assignment of clusters. The multilevel fully interacted model performed well with dissimilar blocks and unbalanced treatment assignment, and instead was most biased when there was a combination of correlation between the number of clusters and the treatment effect, zero cluster-level ICC, and substantial variation in cluster sizes.

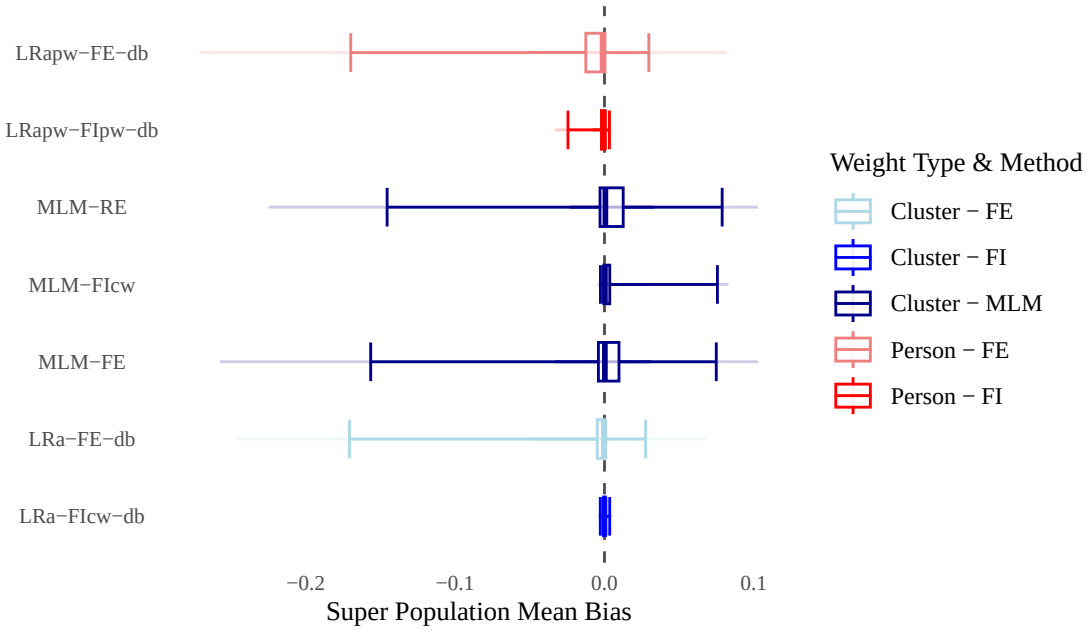


Figure 1: Boxplot of superpopulation mean bias against an estimators targeted estimand across the different scenarios by method. The faint horizontal lines go out to the full range, the bars are the 1% to 99%, and the boxes are the 10% to 90%.

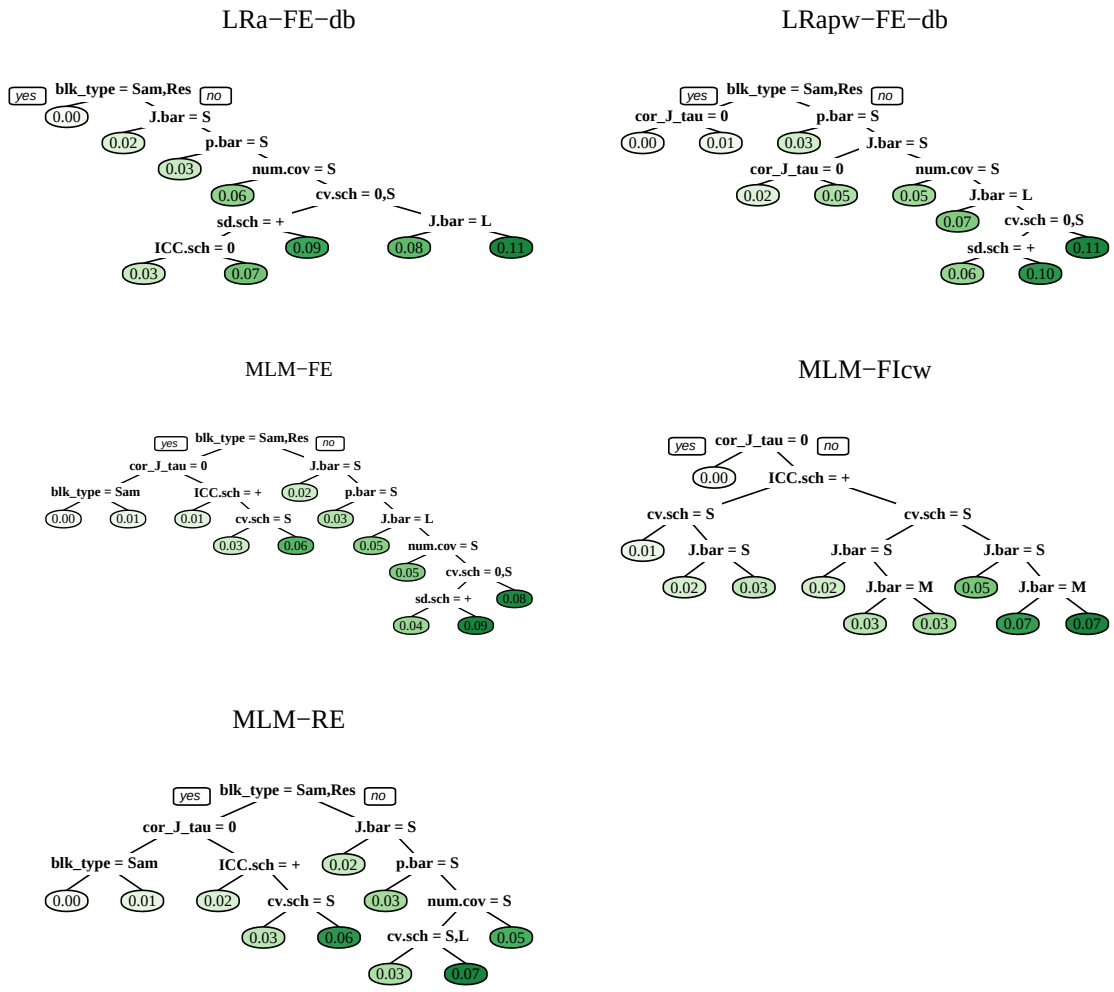


Figure 2: Regression tree by method which predicts the magnitude of the mean bias using different data generating parameters

5 Are any of the estimators generally more precise than others? When?

In general, all the different estimators have comparable precision (Figure 3). Excluding the multilevel estimators, within a given scenario and targeted estimand (person-weighted vs. cluster-weighted), all the estimators have standard errors that are at most 10% larger than the median across methods. The multilevel estimators become more precise than the other estimators when there is a combination of large variation in cluster size and zero school ICC; otherwise, their precision is close to that of the other estimators (Figure 4).

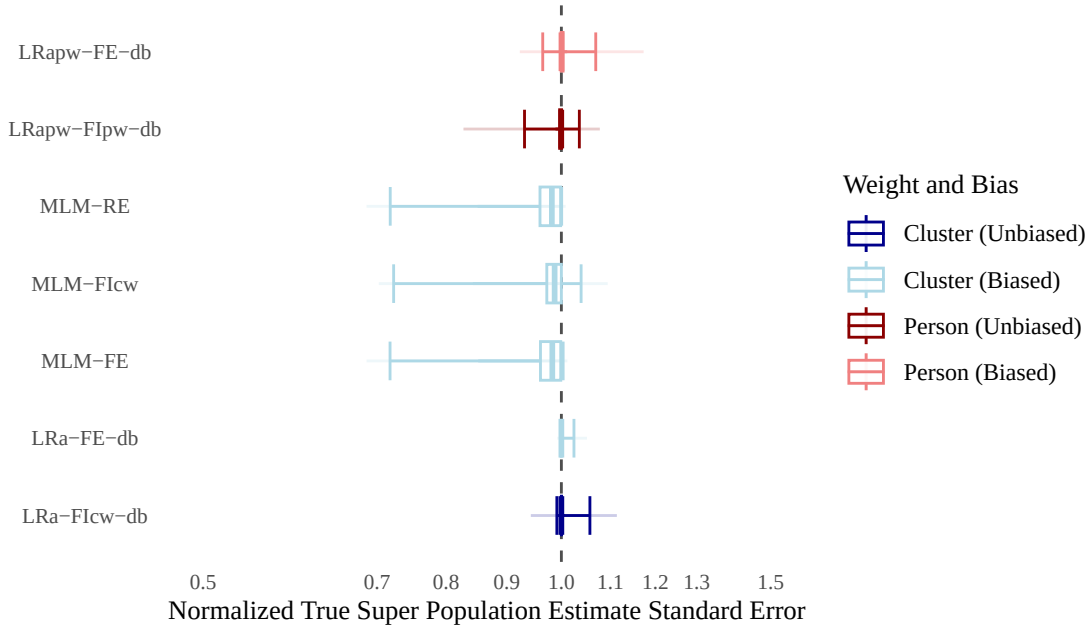


Figure 3: Boxplot of normalized estimate true standard errors across the different scenarios by method. The normalized true standard error is the true standard error for a given method divided by the median true standard error across methods within a given simulation scenario. Therefore each data point is adjusted for

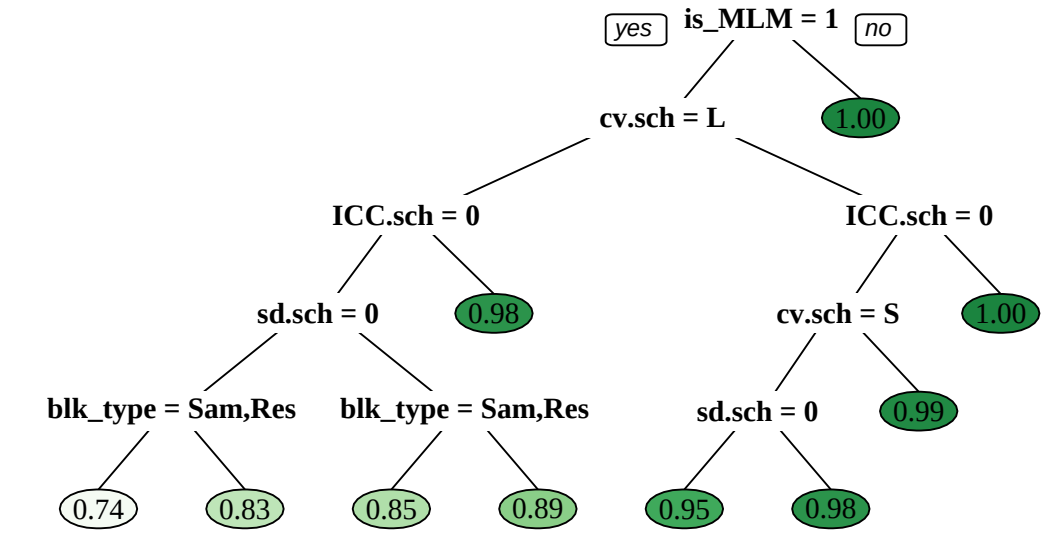


Figure 4: Regression tree by method which predicts the normalized true estimate standard error using different data generating parameters

6 Which estimators produce the best estimates of the treatment effect standard error?

Overall, across the simulations, the standard error estimates for fully interacted estimators, except the fully interacted MLM, are too small (Figure 5). However, across all simulations, the magnitude of the underestimate for the cluster-weighted estimate is very small. For the person-weighted estimators, the standard errors are only substantially underestimated when there is large variation in cluster size combined with either a small number of clusters or large variation across blocks. The only caveat to these results is that almost all the fully interacted estimators are undefined in cases where there are only two clusters in a block. The fully interacted CRVE estimators do have defined standard error estimates in these cases, but these estimates are extremely poor and these estimators should not be used under these circumstances (Figure 6).

On the other hand, the estimated standard errors for the other estimators are on average systematically too high across the simulations. For the MLM, the amount of overestimation is small except when there is a confluence of an unbalanced sample, zero cluster-level ICC, and cross-cluster treatment effect variance (Figure 7). For the fixed effect estimators, the standard error estimates were overestimated by at least 15% in simulations with very dissimilar blocks and no cluster-level ICC (Figure 8).

Consistent with the standard errors being generally well estimated across simulation conditions and estimators, the coverage rate is also close to 95% across simulation conditions and estimators (Figure 9). For the MLM models, sometimes a confluence of multiple simulation conditions (i.e., no variation in cluster size, no cluster-level ICC, and at least three other factors) led the coverage to fall below 90% (Figure 10). Otherwise, there were generally no issues with confidence interval coverage.

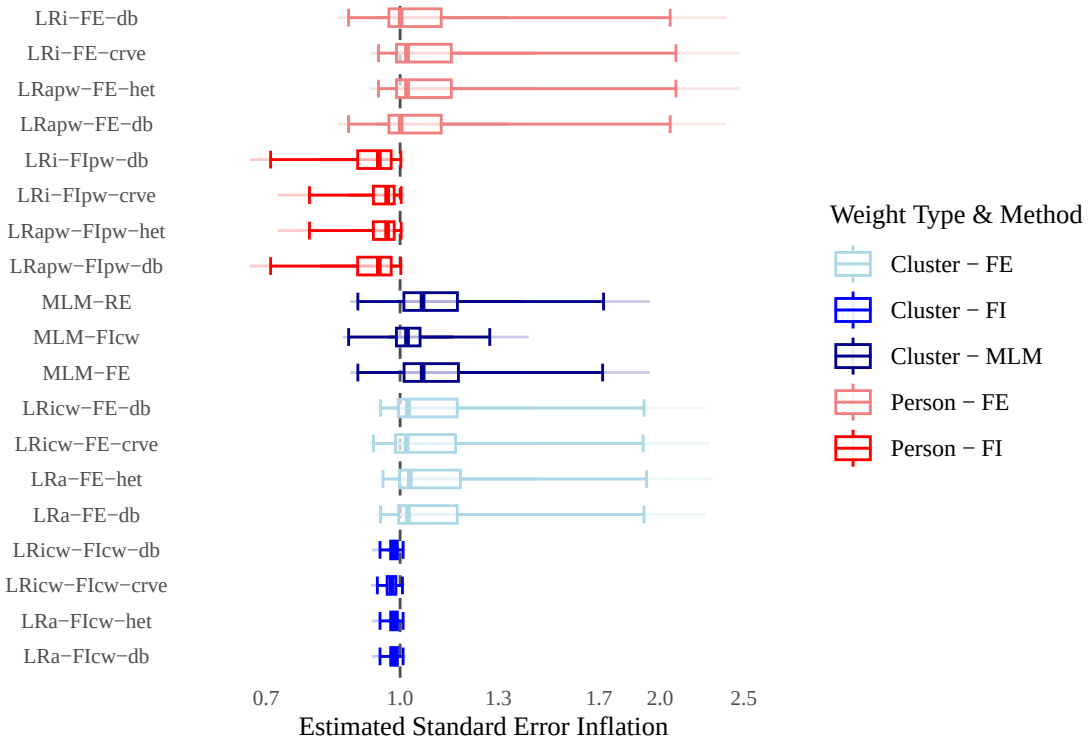


Figure 5: Boxplot of estimated standard error inflation across different simulation scenarios by method. Standard error inflation is measured as the ratio of the estimated standard error over the true super population standard error. Values above 1 indicate overestimation, values below 1 indicate underestimation. The faint horizontal lines go out to the full range, the bars are the 1% to 99%, and the boxes are the 10% to 90%.

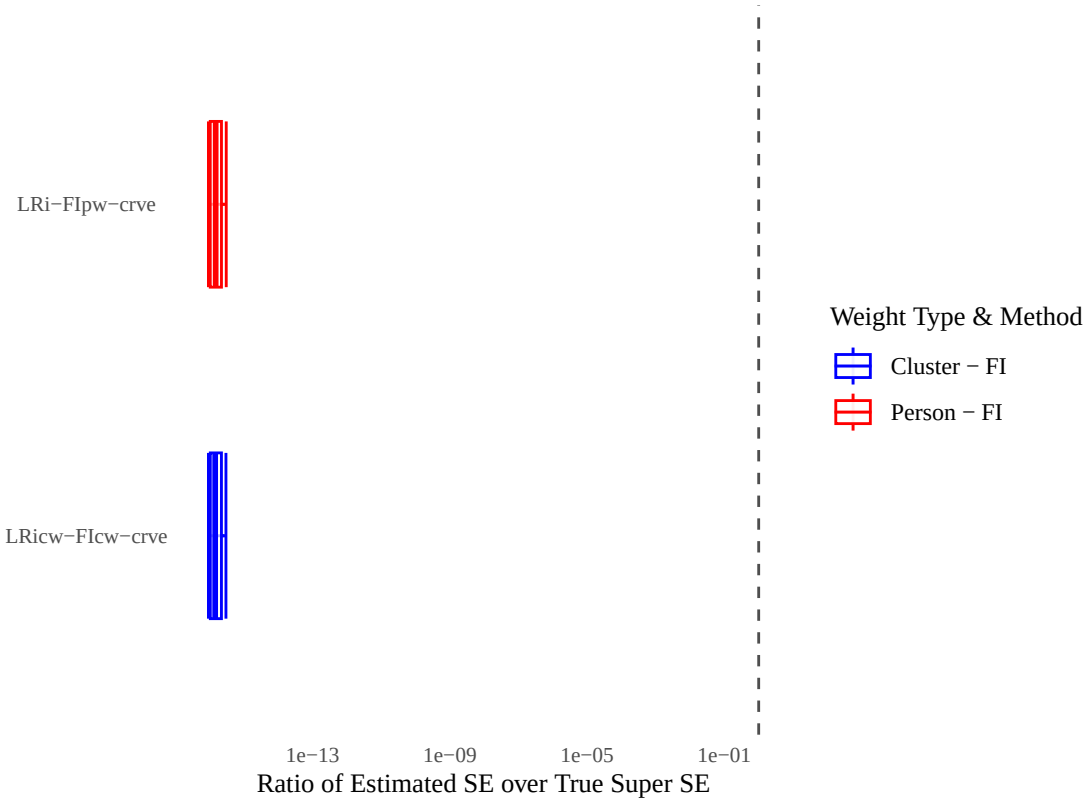


Figure 6: Boxplot of estimated standard error inflation for matched studies for the fully interacted CRVE estimato Standard error inflation is measured as the ratio of the estimated standard error over the true super population standard error. Values above 1 indicate overestimation, values below 1 indicate underestimation. The faint horizon lines go out to the full range, the bars are the 1% to 99%, and the boxes are the 10% to 90%.

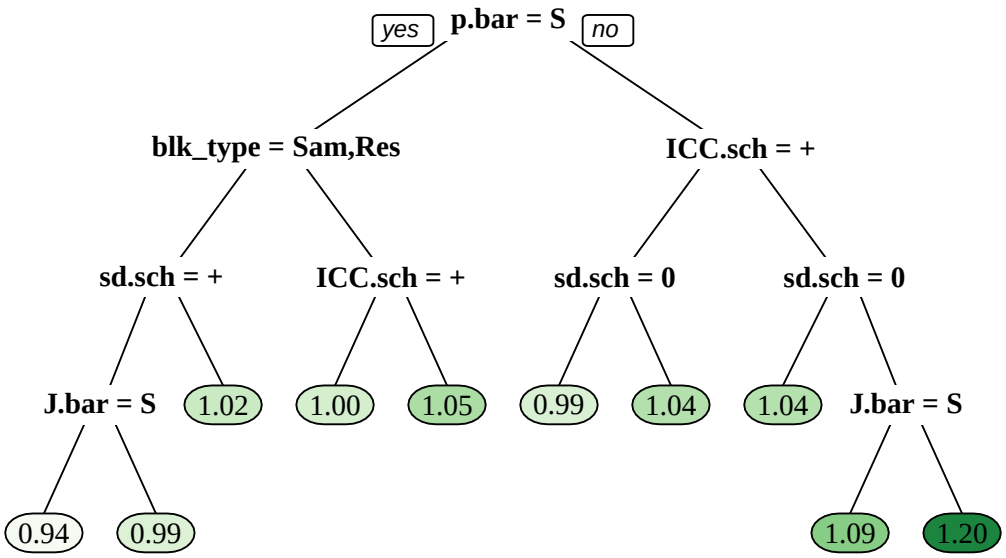


Figure 7: Regression tree which predicts the standard error inflation using different data generating parameters for the multi−level model methods

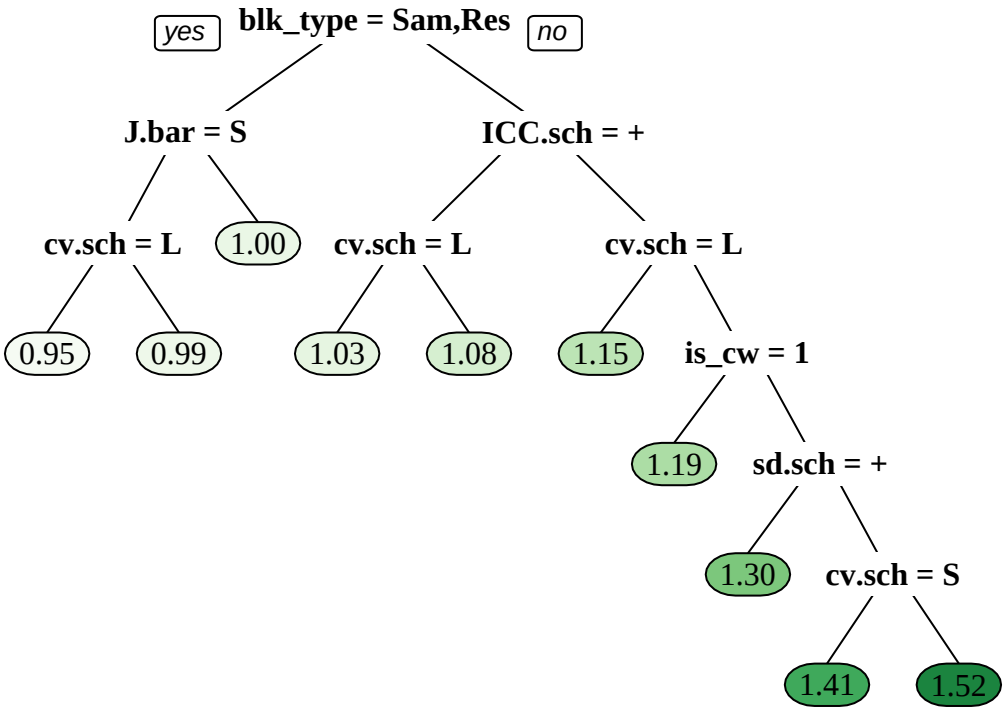


Figure 8: Regression tree which predicts the standard error inflation using different data generating parameters for the fixed effect methods

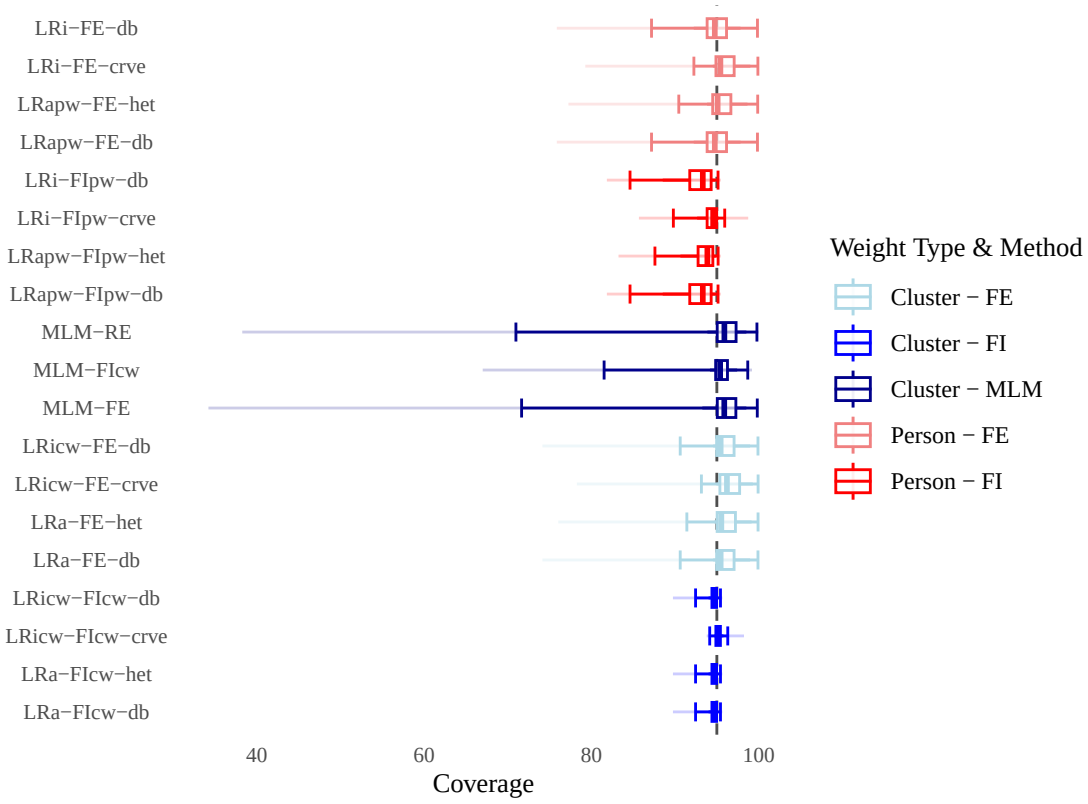


Figure 9: Boxplot of coverage rates across different scenarios by method. The horizontal line indicates the target 95% coverage rate. The faint horizontal lines go out to the full range, the bars are the 1% to 99%, and the boxes are the 10% to 90%.

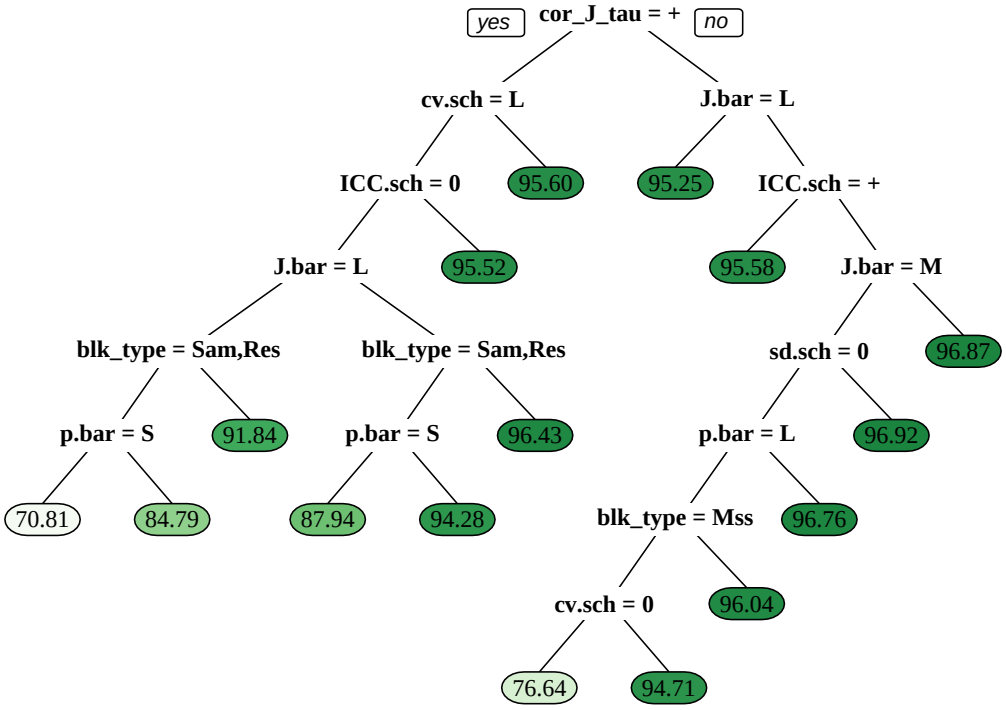


Figure 10: Regression tree which predicts the coverage rates using different data generating parameters for multi−level model methods

7 Do any of the estimated standard errors target finite versus super population uncertainty?

The simulations demonstrate that the standard errors are targeting the super-population standard errors. The true super-population standard errors are only noticeably different from the true finite population standard errors when the cluster-level ICC is zero and there is cross-cluster treatment effect variance (Figure 11). Looking at the subset of simulations where that is true shows that the standard error inflation is consistently closer to one for the true super-population standard errors than for the true finite population standard errors (Figure 12).

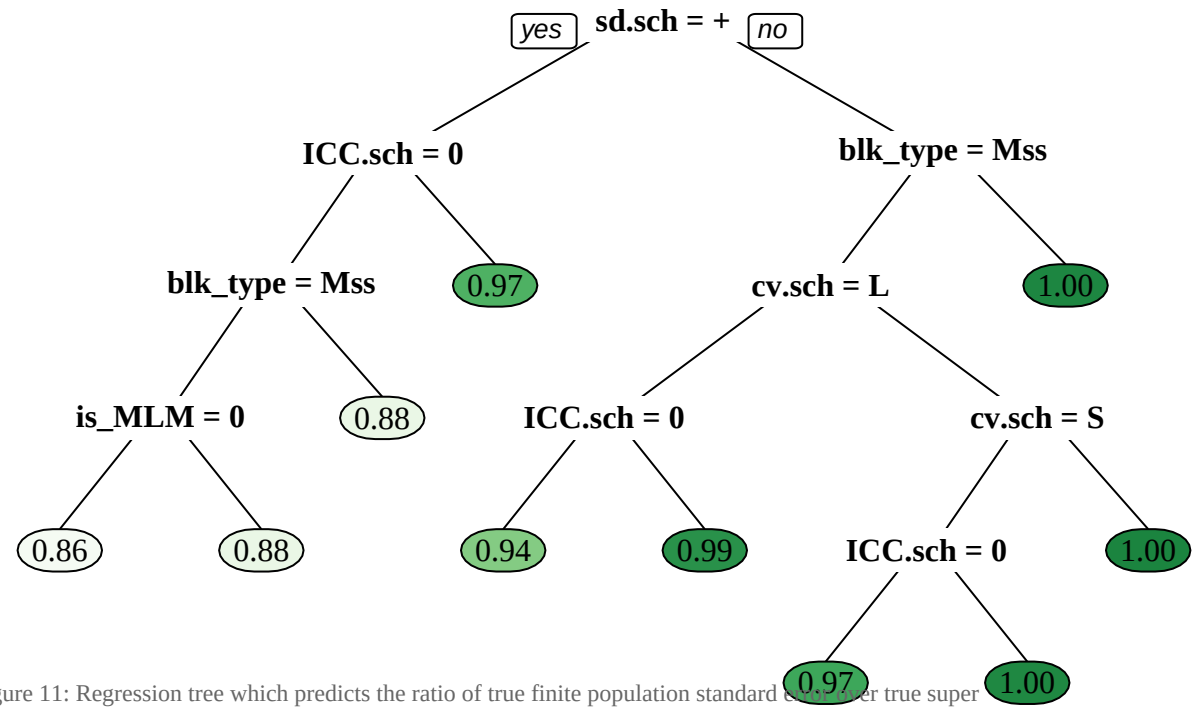


Figure 11: Regression tree which predicts the ratio of true finite population standard error over true super population standard error using different data generating parameters. These two quantities are the same when this ratio is one.

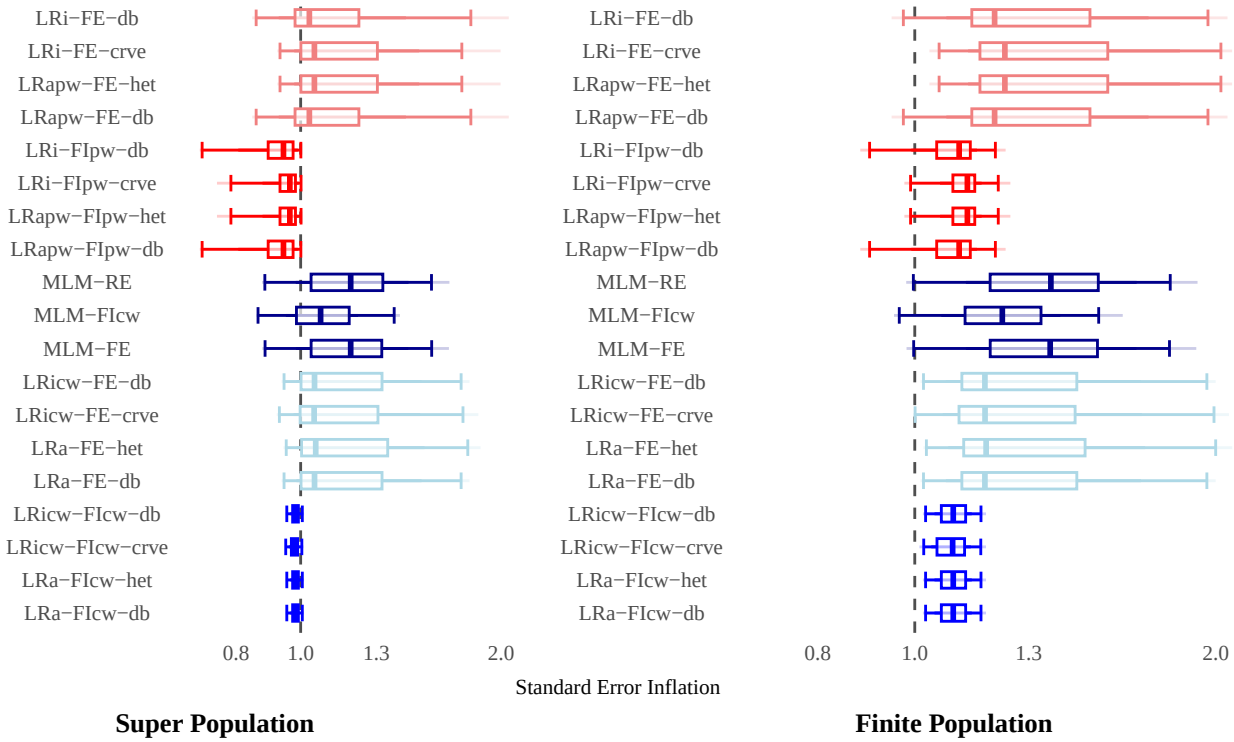


Figure 12: Boxplot of estimated standard error inflation for across simulation scenarios with no cluster−level ICC and non−zero treatment effect heterogeneity by method, comparing againt the true super population and true finite population standard errors. Standard error inflation is measured as the ratio of the estimated standard error over the true standard error. The faint horizontal lines go out to the full range, the bars are the 1% to 99%, and the boxes are the 10% to 90%.

8 Are problems caused by low degrees of freedom due to estimate instability or bias?

Simulations show that the problems with estimates in studies with low degrees of freedom are driven by estimate instability and not estimate bias. We see that as the degrees of freedom get smaller, the estimate instability gets larger (Figure 13). On the other hand, we see no relationship between mean bias and degrees of freedom (Figure 14).

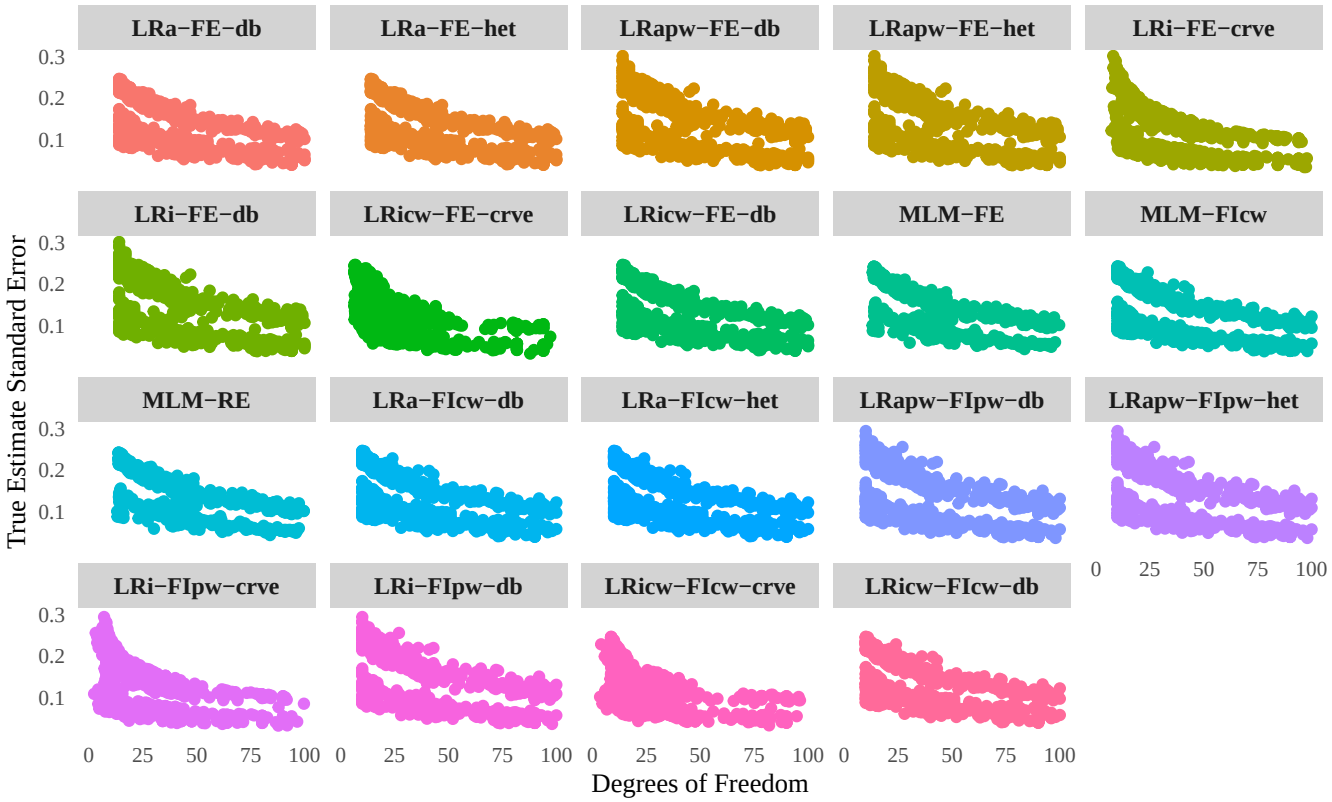


Figure 13: Scatter plot of true estimate standard error versus degrees of freedom by method. Each panel shows the relationship for a different estimation method.

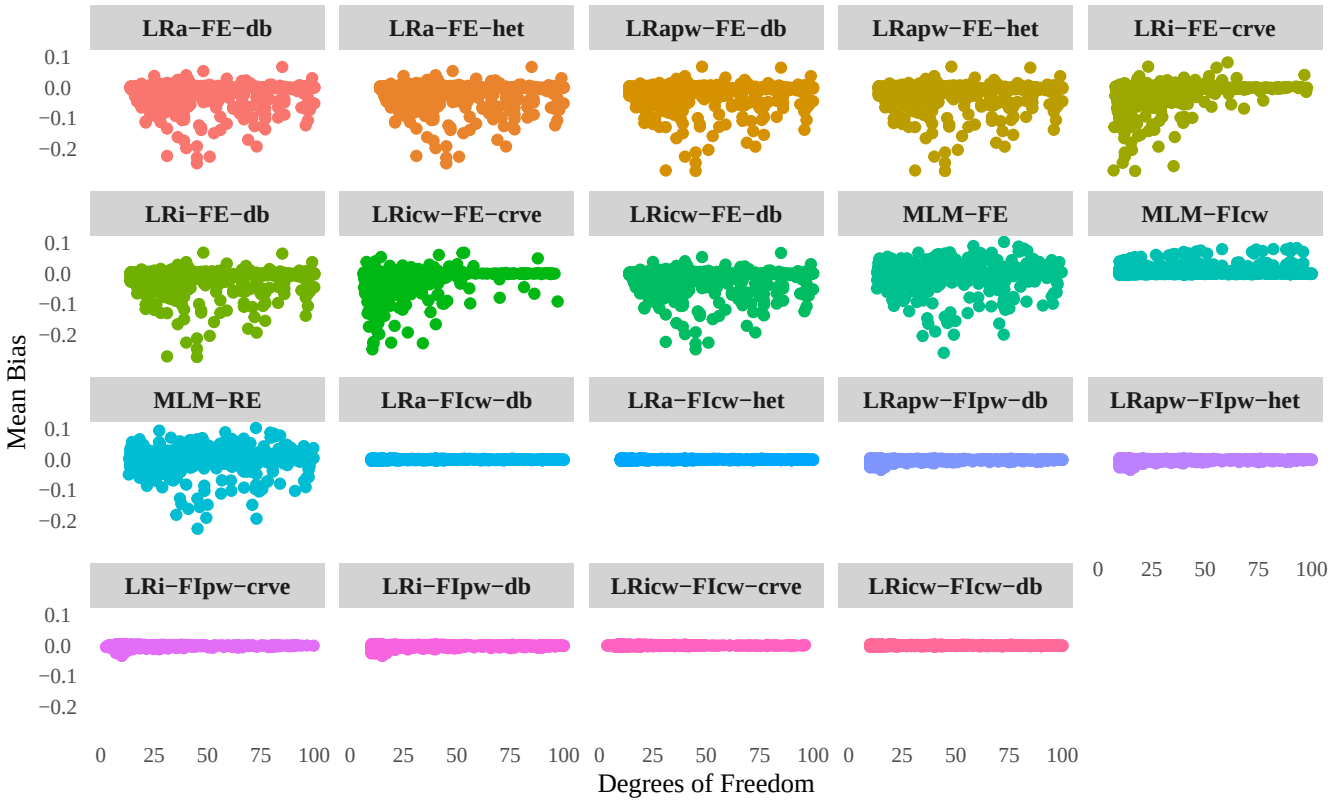


Figure 14: Scatter plot of mean bias versus degrees of freedom by method. Each panel shows the relationship for a different estimation method.